Marc meeting notes

Meeting date: TBD. Created: 2026-10-27.

Starting point

Follow-up to the October 26 meeting, Section 5. Proceed provisionally with liquid-volume-weighted prices and without firm fixed effects. The agreed direction is to select a national firm-month demand specification with a reasonable σ to continue to the supply side, while returning later to the negative price–tax relationship. Gandhi–Houde instruments are a possible future exercise; tax weighting remains to be specified.

1. Sales-weighted tax instruments in the national firm-month grid

Question

Does replacing geographic population weights with sales weights in the tax instrument change which specification families have $0 < \sigma < 1$?

Specification and changes

The previous tax instrument was population-weighted over each brand's operating states, rather than an unweighted UPC average. The new instrument is `Tax_sales_it = sum_s(q_ist * tax_st) / sum_s(q_ist)`, where q is the brand's same-month eligible liquid volume in each state. This uses liquid-volume weights. Taxes retain their existing real-dollar/mL definition and closed-system tax series.

Reran **1,776 specifications with each tax definition**: 1,344 with exogenous quality and 432 with endogenous quality. The exogenous-quality grid includes the original 48 designs. Four control groups (B: nicotine; S: plus device size; V: plus log UPC variety; SV: both), with/without firm FE, both price measures, three market sizes, and the original advisor instrument sets are retained. Nonlag sets run on 528 complete-size rows and again on the 524-row common-lag sample; lag sets and endogenous-quality models use 524 rows. Standard errors remain clustered by month, with no month FE. Only the tax instrument changes within each matched comparison.

Findings

Sales-weighted taxes increase admissible σ point estimates from 357/1,776 (20.1%) to 707/1,776 (39.8%). However, admissible σ combined with negative α increases only from 354 to 374 (19.9% to 21.1% of all runs). Of the new 707 qualifying runs, 333 have nonnegative α . The σ improvement therefore does not imply a comparable improvement in economically plausible demand slopes.

Price / firm FE	Population tax: qualifying / total	Sales tax: qualifying / total	Sales tax: also negative α
Volume weighted, no firm FE	232 / 444 (52.3%)	235 / 444 (52.9%)	149 / 444
Volume weighted, firm FE	83 / 444 (18.7%)	174 / 444 (39.2%)	167 / 444
Equal-UPC, no firm FE	35 / 444 (7.9%)	142 / 444 (32.0%)	16 / 444
Equal-UPC, firm FE	7 / 444 (1.6%)	156 / 444 (35.1%)	42 / 444

Going in the direction of **volume-weighted prices and no firm FE**, per the last meeting:

- Tax + annual incumbent count gives qualifying σ in **24/24** runs, with negative α in **20/24**. Tax + annual and monthly incumbent counts also gives **24/24**, with negative α in **20/24**.
- Tax + annual count + lag quality gives **24/24**, with negative α in **16/24**. Adding monthly count to this set gives **24/24**, with negative α in **20/24**.
- Tax + monthly incumbent count gives **21/24**, with negative α in **17/24**. Tax + monthly count + lag gives

16/24, with negative α in **12/24**.
- Adding annual count to the original national or geographic active-brand-count sets gives **24/24** qualifying sigma, but only **8/24** and **11/24**, respectively, also have negative α .
- Tax + either original active-brand count alone still gives **0/24** qualifying σ in this price/FE family. Across the entire grid, the original 48 designs also have **0/48** qualifying σ with the new tax weights.

These 24-run instrument families combine control groups, market sizes, and the applicable sample/quality comparisons; they are not independent replications. All listed instrument sets include tax. See the focused instrument-family table for every set.

Other patterns across the sales-tax grid:

- **Controls:** SV has 203/444 qualifying runs; V 192/444; S 160/444; B 152/444. Requiring negative α gives 97, 81, 104, and 92, respectively, so the ranking changes.
- **Market size:** monthly intensity gives 305/592 qualifying runs, annual 231/592, constant intensity 171/592. Requiring negative α gives 98, 105, and 171. Monthly intensity's larger σ count does not carry over to the joint sign criterion.
- **Quality:** exogenous-quality models qualify in 517/1,344 runs (291 also negative α); endogenous-quality models in 190/432 (83 also negative α).
- **Lag quality is not required:** 360/864 nonlag runs and 347/912 lag runs qualify.
- **Matched transitions:** 374 specifications move into (0,1), 24 move out, 333 stay inside, and 1,045 stay outside.

σ levels among qualifying runs

Among the **707 sales-tax runs with $0 < \sigma < 1$**, mean σ is **0.780** and median σ is **0.831**. Under the original population tax weights, the corresponding values were **0.573** and **0.616** among 357 qualifying runs. Each specification receives equal weight, including deliberate sample-matched repetitions; these are conditional descriptive summaries, not pooled estimates.

Price and fixed effects — sales-weighted taxes

Family	Qualifying runs	Mean σ	Median σ
Equal-UPC, firm FE	156	0.791	0.798
Volume weighted, firm FE	174	0.641	0.744
Equal-UPC, no firm FE	142	0.786	0.826
Volume weighted, no firm FE	235	0.871	0.880

Instrument sets — sales-weighted taxes

Family	Qualifying runs	Mean σ	Median σ
Annual	63	0.748	0.813
Annual_Monthly	64	0.750	0.816
Geographic	0	—	—
Geographic+incumbent_annual	64	0.772	0.798
Geographic+incumbent_monthly	29	0.844	0.855
Monthly	52	0.769	0.844
National	0	—	—
National+incumbent_annual	64	0.792	0.839
National+incumbent_monthly	24	0.859	0.883
Annual_Lag	63	0.780	0.838
Annual_Monthly_Lag	65	0.769	0.839
Geographic+a_lag1	0	—	—
Geographic+incumbent_annual+lag	63	0.770	0.799
Geographic+incumbent_monthly+lag	27	0.819	0.830
Lag	1	0.076	0.076
Monthly_Lag	45	0.785	0.859

Family	Qualifying runs	Mean σ	Median σ
National+a_lag1	0	—	—
National+incumbent_annual+lag	59	0.774	0.814
National+incumbent_monthly+lag	24	0.839	0.865

Controls — sales-weighted taxes

Family	Qualifying runs	Mean σ	Median σ
Nicotine only	152	0.809	0.863
Nicotine + size + log variety	203	0.762	0.806
Nicotine + size	160	0.788	0.847
Nicotine + log variety	192	0.769	0.828

Market size — sales-weighted taxes

Family	Qualifying runs	Mean σ	Median σ
Annual intensity	231	0.807	0.832
Constant intensity	171	0.682	0.805
Monthly intensity	305	0.814	0.839

Quality treatment — sales-weighted taxes

Family	Qualifying runs	Mean σ	Median σ
Exogenous	517	0.782	0.831
Endogenous	190	0.775	0.830

Instrument labels above all include tax: Annual/Monthly denote incumbent counts; National/Geographic denote the original active-brand counts; Lag denotes previous-calendar-month quality. Combinations list the additional instruments. Zero qualifying runs have no conditional mean or median.

For the provisional volume-weighted-price/no-FE direction, the 235 qualifying runs have **mean 0.871 and median 0.880**. This summary includes positive price slopes; the CSV also reports mean/median among qualifying runs with negative α separately. The focused instrument-family table now includes mean and median σ for each instrument set within that price/FE family.

Interpretation and next steps

Annual-incumbent-count instruments remain a prominent qualifying family, particularly with volume-weighted prices. Under the provisional no-FE direction, tax + annual count (with or without monthly count) warrants closer examination of coefficients, uncertainty, and identification. It is not yet a selected supply-side specification.

Sales weighting does not resolve the original tax + active-brand-count-only models. Many newly admissible σ estimates have the wrong price sign. Current sales weights can correlate with demand shocks and quantity measurement error; these results do not establish exclusion validity. Strict qualification concerns the σ point estimate only, not its confidence interval, quality coefficient, or instrument strength. Repeated sample comparisons inflate neither independent evidence nor confidence in a family. No σ clipping or canonical replacement was performed.

Files and checks

The population-weighted rerun reproduces the historical **357 qualifying / 354 negative-** α totals. All 48 saved original baseline coefficients and covariances reproduce (maximum discrepancies below 8e-11). All 3,552 fits pass

independent QR coefficient and structural-residual checks (maximum discrepancies below 6e-9); conditional-F validation is retained for two or three endogenous regressors. All coefficient SEs are finite. State sales reconcile to saved national brand-month volume and revenue within 1e-8 relative tolerance; tax weights sum to one and weighted tax lies within operating-state tax bounds. Source hashes are unchanged.

- Complete family breakdown.
- Family counts, all specification results, qualifying sales-tax runs, and matched comparison.
- Detailed coefficients, first stages, conditional F, models, checks, weights, and input hashes: `output/demand_estimation/sal` `TaxIV_geo` is retained as the internal variable name; each mode directory identifies the weighting definition.
- Scripts: `code/3_demand_estimation/23_sales_weighted_tax.R` and `24_sales_tax_family_report.R`. Existing inputs, NCP data, and prior results are preserved.

2. Compare all 24 specifications in the annual-incumbent-count family

Question and reason for shortlisting

Compare all candidates within **volume-weighted prices + no firm FE + sales-weighted tax + annual incumbent count**, before choosing a provisional national demand specification for supply-side work. **This family was shortlisted because it is one of the most successful families by the frequency of admissible σ point estimates:** 24/24 qualify and 20/24 also have negative α . Other families tie on σ qualification. This criterion does not establish instrument validity or select a final specification.

The earlier illustration focused prematurely on both product controls. The full choice set is **four control groups $\times$ three market sizes $\times$ two samples = 24**. There are 12 substantive control/market combinations; the second sample checks sensitivity to dropping four rows.

Common specification

- National incumbent firm-month observations; price is real revenue / eligible liquid volume.
- Price and within-nest log share jointly endogenous. Excluded instruments: brand-specific current liquid-volume-weighted state tax and annual national incumbent count (>7% annual volume share).
- Current nicotine quality is included and treated as exogenous in all runs. B: nicotine only; S: plus mean device size; V: plus log UPC variety; SV: both.
- Intercept, no firm FE, no month FE; SEs clustered by month. Exactly identified (two excluded instruments / two endogenous regressors).
- Available sample: 528 complete-size rows / 132 months. Common-lag sample: 524 rows / 131 months. No lagged-quality instrument is used in these 24 runs; the smaller sample is a comparability check only. All control groups retain the same complete-size sample rule.
- Annual, monthly, and constant intensity use the existing market-size definitions.

All 24 estimates

Entries are coefficient (month-clustered SE). Quality is current delivered nicotine. Every σ point estimate below lies in (0,1).

Available sample: 528 rows

Controls	Market intensity	Price α (SE)	σ (SE)	Quality (SE)
B	annual	-0.0701 (0.0341)	0.8374 (0.2212)	0.0606 (0.0081)
B	monthly	-0.0584 (0.0384)	0.8609 (0.2351)	0.0588 (0.0089)
B — chosen	**constant**	**−0.3336 (0.0391)**	**0.8695 (0.2373)**	**0.1100 (0.0092)**
S	annual	-0.0807 (0.0411)	0.8182 (0.2326)	0.0570 (0.0063)
S	monthly	-0.0659 (0.0459)	0.8475 (0.2475)	0.0562 (0.0068)
S	constant	-0.3906 (0.0540)	0.7671 (0.2535)	0.0908 (0.0072)
V	annual	-0.0025 (0.0389)	0.8039 (0.1792)	0.0608 (0.0064)
V	monthly	0.0107 (0.0426)	0.8267 (0.1945)	0.0589 (0.0073)

Controls	Market intensity	Price α (SE)	σ (SE)	Quality (SE)
V	constant	-0.3237 (0.0409)	0.8646 (0.2383)	0.1100 (0.0090)
SV	annual	-0.0044 (0.0393)	0.7842 (0.1830)	0.0577 (0.0049)
SV	monthly	0.0095 (0.0434)	0.8139 (0.2012)	0.0569 (0.0056)
SV	constant	-0.3352 (0.0394)	0.7424 (0.2251)	0.0913 (0.0065)

Common-lag sample: 524 rows

Controls	Market intensity	Price α (SE)	σ (SE)	Quality (SE)
B	annual	-0.0676 (0.0349)	0.8594 (0.2251)	0.0596 (0.0083)
B	monthly	-0.0559 (0.0394)	0.8832 (0.2396)	0.0577 (0.0092)
B	constant	-0.3367 (0.0398)	0.8452 (0.2392)	0.1112 (0.0094)
S	annual	-0.0780 (0.0421)	0.8422 (0.2361)	0.0561 (0.0064)
S	monthly	-0.0630 (0.0472)	0.8715 (0.2516)	0.0553 (0.0070)
S	constant	-0.3954 (0.0552)	0.7479 (0.2570)	0.0914 (0.0074)
V	annual	-0.0019 (0.0390)	0.8192 (0.1830)	0.0601 (0.0066)
V	monthly	0.0115 (0.0429)	0.8420 (0.1990)	0.0582 (0.0075)
V	constant	-0.3237 (0.0410)	0.8372 (0.2395)	0.1113 (0.0092)
SV	annual	-0.0040 (0.0397)	0.8008 (0.1871)	0.0572 (0.0051)
SV	monthly	0.0102 (0.0438)	0.8306 (0.2060)	0.0564 (0.0059)
SV	constant	-0.3375 (0.0399)	0.7156 (0.2267)	0.0923 (0.0067)

Patterns and qualifications

- σ**:** all 24 point estimates qualify, ranging from 0.716 to 0.883. However, **0/24 conventional 95% intervals lie wholly within (0,1)**; all extend above 1.
- **Price:** 20/24 α estimates are negative. The four positive estimates are the monthly-intensity V and SV runs in both samples. Only 9/24 price intervals are entirely below zero: the eight constant-intensity runs and the available-sample B/annual run. All other annual/monthly price intervals include zero; B/annual loses that exclusion of zero on the common-lag sample.
- **Variety control:** with annual or monthly intensity, including log UPC variety (V or SV) shifts α toward zero or slightly positive. Without variety (B or S), these price estimates are more negative, but remain imprecise. This sensitivity does not by itself justify dropping variety.
- **Market size:** constant intensity gives substantially more negative α in every control group (approximately -0.32 to -0.40). Do not adopt this market-size assumption solely to obtain the desired price sign.
- **Quality:** every quality point estimate is positive; all 24 conventional 95% quality intervals are above zero.
- **Sample sensitivity:** the smaller sample leaves the main sign patterns unchanged. The B/annual price interval narrowly excludes zero in the available sample but includes it in the smaller sample. The smaller sample contains no additional lag instrument and is not independent evidence.

First-stage diagnostics by control group and sample

First stages are identical across the three market sizes within a control/sample combination. Tax coefficients are estimate (SE); F values below are SW conditional F for each endogenous regressor. Full instrument and control coefficients, ordinary joint F, and conditional-F details are exported.

Sample	Controls	Tax $\rightarrow$ price (SE)	Price SW F	Share SW F
available	B	-7.505 (1.372)	79.90	172.85
available	S	-6.137 (1.377)	49.37	129.34
available	V	-6.660 (1.325)	59.21	136.30
available	SV	-6.006 (1.342)	60.58	90.20
common_lag	B	-7.178 (1.362)	74.57	165.06

Sample	Controls	Tax → price (SE)	Price SW F	Share SW F
common_lag	S	-5.860 (1.366)	46.56	122.89
common_lag	V	-6.455 (1.323)	57.13	129.24
common_lag	SV	-5.778 (1.337)	57.91	84.59

Tax–price slopes remain negative in all eight designs. Conditional F does not establish instrument exclusion or calibrated weak-IV-robust inference. Exact identification leaves no overidentifying restrictions to test. Both sales-based tax weights and sales-share-threshold counts require an exclusion argument.

Gandhi–Houde instruments: first-pass comparison before choosing a specification

The first trial of G–H-style local-neighbor and quadratic-distance instruments did **not clearly improve the current no-firm-FE specification**. All runs used volume-weighted prices; where tax was included it used sales weights. Current characteristics were treated as exogenous.

For nicotine-only controls and constant-intensity market size on the same 528 observations:

Excluded instruments	Alpha	Sigma
Tax + annual incumbent count: baseline	−0.334	0.869
Tax + local nicotine differentiation	−0.339	1.740
Tax + quadratic nicotine differentiation	−0.322	−0.887
Tax + both differentiation measures	−0.336	1.213
Baseline + local differentiation	−0.365	0.935
Baseline + both differentiation measures	−0.358	0.973

- Replacing annual incumbent count with these nicotine differentiation instruments did not yield admissible sigma in the closest comparison.
- Supplementing the baseline sometimes retained admissibility but moved sigma closer to one. This alone is not evidence of improved identification or a better demand model.
- Size-based instruments require dropping all 2018 months to retain a complete rival characteristic set. Those estimates must be compared against baselines refitted on that same smaller sample.
- The first pass retained 132 numerically validated fits; 12 planned fits were underidentified and six were numerically unstable. Differentiation instruments based only on nicotine and without tax/count were extremely weak in this application.
- This finding concerns the tested aggregate-brand constructions, not a general rejection of G–H instruments. A broader matched-sample trial is being examined. The specification and recovered costs used for the pricing fixed point remain unchanged pending discussion.

Results: first-pass findings. No new demand specification has been adopted on the basis of this exercise.

Expanded G–H sensitivity trial

The extended trial tests all seven nonempty combinations of nicotine, device size, and log variety; local neighborhoods of 0.5, 1, and 2 pooled standard deviations of within-month pairwise differences; local signed-difference and quadratic cross-characteristic interactions; replacing versus supplementing annual/monthly count instruments; and, for nicotine, including the passive aggregate fringe among rival alternatives. Controls include the own characteristics used to construct instruments. All runs retain volume-weighted prices, no firm FE, and exogenous current characteristics. Baselines are refitted on exactly matched samples; size-based families drop all 2018 months for complete rival coverage.

- **2,129 fits pass numerical validation**, with 646 admissible sigma estimates and 561 also having negative alpha. These counts include deliberate bandwidth-invariant repetitions and baseline fits across different control/sample families, not independent successes.
- An additional 216 market-specific fits are underidentified and 31 are numerically unstable; they are recorded separately, not interpreted as trustworthy estimates.

- Independent matrix calculations verify local, quadratic, and interaction instruments across 3,024 variant/characteristic/month groups, agreeing within 9e-14. Retained coefficient and residual checks pass; matched baselines and input hashes are unchanged.

Closest comparisons to the chosen nicotine-only, constant-intensity specification, all using the same 528 rows:

Instruments / construction	Alpha	Sigma (SE)
Current baseline: tax + annual incumbent count	−0.334	0.869 (0.237)
Tax + local, incumbent rivals, width 2 SD	−0.334	0.900 (0.161)
Tax + local interactions, incumbent rivals, width 2 SD	−0.291	0.955 (0.141)
Baseline + local, incumbents and fringe, width 0.5 SD	−0.317	0.773 (0.221)

The expanded exercise **qualifies the initial finding**: some alternative constructions give admissible sigma and smaller SEs, so the first pass does not establish that G–H instruments cannot help. However, results are sensitive to neighborhood and rival definitions: tax + incumbent-local gives sigma 4.119 at width 0.5 SD, 1.740 at 1 SD, and 0.900 at 2 SD. All sigma intervals in the table still extend above one. A lower SE or admissible point estimate alone does not establish instrument validity or a superior demand specification.

Retain the existing specification as the provisional pricing baseline while discussing these sensitivity results. No replacement demand fit, cost recovery, or pricing equilibrium has been adopted. Characteristic positioning/assortment exogeneity and the sales-weighted-tax exclusion remain assumptions; the extension does not resolve them.

Extended findings, all specifications and first-stage strength, matched comparisons, and chosen-specification neighborhood. Scripts: `28_extended_gandhi_houde.R` and `29_validate_extended_gh.R` in `code/3_demand_estimation/`. Full coefficient/SE and validation tables are saved by variant; no full fitted-model objects are retained.

Chosen provisional specification and rationale

Chosen for the next stage: nicotine-only (B), constant-intensity market size, available sample (528 observations / 132 months), volume-weighted prices, no firm FE, and sales-weighted tax + annual incumbent count instruments. The corresponding row is bold above.

This specification was chosen because ** α is negative and not too small in magnitude: −0.334 (SE 0.039)**, and it belongs to one of the most consistently successful families by admissible σ:** 24/24 family runs have $0 < \sigma < 1$, and 20/24 also have negative α **. Thus 20/24 is the count satisfying both the σ and price-sign criteria, not the sigma-only count. The chosen σ is** 0.869 (SE 0.237)** and the quality slope is **0.110 (SE 0.009)**. This is a pragmatic provisional choice following the last meeting, not evidence that this specification or market-size assumption is uniquely correct.

Nested-logit equation for the chosen specification

Using the notation in the previous presented troubleshooting notes:

$$\log(s_{it}/s_{0t}) = c + \alpha p_{it} + \beta_a a_{it} + \sigma \log(s_{it}/s_{gt}) + \xi_{it}.$$

i indexes national incumbent brands and t months. p_{it} is the liquid-volume-weighted real price per mL, a_{it} is delivered nicotine concentration, and ξ_{it} is the unobserved demand term. Intercept c is included.

The shares use constant-intensity market size:

$$M_t = \bar{k}\,\text{Adults}_t, \qquad s_{it} = q_{it}/M_t, \qquad s_{gt} = Q_t/M_t, \qquad s_{0t} = 1 - Q_t/M_t,$$

where q_{it} is eligible brand liquid volume, Q_t sums eligible incumbent and fringe volumes, and $\bar{k}$ is the full-period mean of the monthly NCP purchasing-household consumption-intensity series. Constant intensity does not mean constant market size: adult population varies over time.

The estimated equation, with coefficients rounded, is:

$$\log(s_{it}/s_{0t}) = -6.8938 - 0.3336\, p_{it} + 0.1100\, a_{it} + 0.8695 \log(s_{it}/s_{gt}) + \widehat{\xi}_{it}.$$

Price and within-nest log share are jointly instrumented by **sales-weighted state tax and annual national incumbent count**. Current quality is treated as exogenous. The two excluded instruments identify the two endogenous regressors exactly; standard errors are clustered by month.

The negative conditional tax–price relationship remains unresolved. σ 's conventional 95% interval extends above 1, and the constant-intensity market-size assumption still requires substantive support.

Files

All 24 specifications with confidence intervals and diagnostics, all coefficients, and complete first stages. These supersede the six-run illustration as the main shortlist comparison. Extraction script: `code/3_demand_estimation/26_compare_annua` no regressions were rerun.

For next time, attempt Gandhi–Houde instruments, given no firm FE.

3. Supply-side setup: cost recovery and fringe demand

Decisions before coding

Start by recovering demand residuals and marginal costs at observed prices, then implement the static pricing fixed point manually and test observed-market replication. Do not use packaged black-box equilibrium solvers. The planned code folder is `code/4_static_pricing_subgame/`. Treat each brand as a vertically integrated pricing firm, as agreed: it chooses its representative retail price and receives the corresponding retail revenue. Recovered marginal costs are effective integrated-channel costs; operating profits are integrated-channel profits, rather than separately identified manufacturer profits. No separate wholesale-price or retailer-margin stage is modeled. This is the initial approximation for the static subgame and must remain explicit when interpreting later entry/exit results.

Fringe utility includes unobserved quality

Decision: allow fringe mean utility to include unobserved quality, rather than restricting it to measured nicotine quality alone. In the signed- α convention, the full mean utility is

$$\delta_{F,t} = c + \beta_a a_{F,t} + \alpha p_{F,t} + \xi_{F,t}, \qquad \alpha < 0.$$

The fringe is one aggregate alternative in the same e-cigarette nest. For the initial observed-market replication, calibrate its mean utility from observed shares:

$$\delta^{\text{obs}}_{F,t} = \log(s_{F,t}/s_{0t}) - \sigma \log(s_{F,t}/s_{gt}).$$

Given consistently defined observed fringe price and quality, recover the implied unobserved component as

$$\xi_{F,t} = \delta^{\text{obs}}_{F,t} - c - \beta_a a_{F,t} - \alpha p_{F,t}.$$

For the first static solve, it is sufficient to use calibrated total fringe utility; separating price, quality, and the residual requires consistent fringe aggregates. Calibration matches the observed aggregate share but does not estimate a separate fringe demand equation or validate the assumed aggregation.

Fringe utility may vary across months. Hold it fixed within each month's incumbent pricing game: the fringe remains passive, with no strategic price choice. Its share nevertheless responds to incumbent prices through the demand denominator. This changes the PDF restriction $\delta_{F,t} = \beta_a a_{F,t}$; it does not require a strategic fringe.

Current quality and demand shocks are held fixed during each static solve. Their Markov transitions are deferred until the dynamic continuation-value/entry-exit stage; no new transition assumptions have been selected here.

Revised supply documents

Reconstructed editable LaTeX sources for the conference model and solution appendix, and regenerated their PDFs. Both now use signed α in `+ $\alpha$ * price`, include the demand intercept and calibrated fringe residual, and update derivatives, markups, cost recovery, and manual fixed-point equations consistently. Original PDFs are preserved in `docs/source_archive/pre_reconstruction/`; the model retains its original chart and historical concentration table. Unresolved dynamic-cap and private-cost assumptions are flagged explicitly. Algebra and rendered pages were checked; no pricing solver was implemented. Rebuild with `bash smoke_firm_dir/docs/build_supply_documents.sh`.

Reconstruction audit completed

A section-by-section audit compared both reconstructed supply documents with their archived originals. It restored historical tax/count definitions, missing-size exclusions, absorbing exit, the ownership-exit sample statement, and explicit dynamic iteration details that had been compressed. The concentration table matches all 11 original rows, and the chart uses the original PDF figure. Intended changes (signed α , calibrated fringe utility, current provisional specification) and corrections to contradictory draft statements are documented separately in the audit record. Both PDFs were regenerated and checked. These remain revised reconstructions rather than verbatim copies; the original PDFs remain unchanged.

Observed-market recovery completed

Implemented the first supply-side recovery step in `code/4_static_pricing_subgame/`, reading the full-precision chosen demand coefficients. No price fixed point has been solved yet.

The demand fit uses **528 observations**, but complete observed markets have **540 incumbent firm-month rows across 132 months**. Keep the 12 BLU 2018 rows excluded for missing device size as competitors; invert their observed shares using the chosen parameters and flag them as outside the estimation sample. This preserves observed inside volumes and the full competitive roster without refitting demand or imputing size. Their residuals are out-of-estimation-sample inversions, not residuals from rows used in IV estimation.

For each incumbent, recover

$$\widehat{\xi}_{it} = \log(s_{it}/s_{0t}) - \widehat{c} - \widehat{\alpha} p_{it} - \widehat{\beta}_a a_{it} - \widehat{\sigma}\log(s_{it}/s_{gt}),$$

$$A_{it} = \frac{1}{1-\widehat{\sigma}} - \frac{\widehat{\sigma}}{1-\widehat{\sigma}} s_{i|g,t} - s_{it}, \qquad \widehat{\mu}_{it} = -\frac{1}{\widehat{\alpha} A_{it}}, \qquad \widehat{mc}_{it} = p_{it} - \widehat{\mu}_{it}.$$

Variable profit is $(p_{it} - \widehat{mc}_{it})q_{it}$, before unspecified fixed operating costs. No extra tax is subtracted from recovered costs; their interpretation follows the observed retail-price convention. The fringe's existing price is real fringe revenue / fringe volume, and its quality is the existing equal-UPC delivered-nicotine measure. Calibrate total fringe utility from shares and recover its residual decomposition as agreed above; no fringe pricing FOC or fringe cost is inferred.

Initial findings: no negative marginal costs among all 540 rows. On the **528-row estimation sample**:

Quantity	Result
Median retail price	\$4.161/mL
Median recovered markup	\$0.459/mL
Median effective marginal cost	\$3.701/mL
Marginal-cost range	\$0.651–\$9.398/mL
Total variable profit, 2013–2023 sample	\$535.33 million

The extra 12 BLU rows contribute $3.84 million in variable profit, giving **$539.16 million across the full 540-row observed roster**. These are totals across eligible observed sales over the full period, not annual profit, manufacturer profit, or profit net of fixed costs. Medians are computed separately, so median price need not equal median cost plus median markup.

Checks: independently reproduced the chosen IV coefficients and residuals using QR 2SLS. Full incumbent volumes and revenue reconcile to saved panels. Reconstructed incumbent/fringe/outside shares match observed shares to numerical precision; maximum relative quantity error is below 2e-14. Scaled observed-price FOCs are below 2e-15. Finite-difference demand checks pass for all 540 incumbent price perturbations, including cross-price and passive-fringe responses (maximum derivative discrepancy below 2e-12). The sigma-zero logit limit, extreme-utility stability, parameter guards, and unchanged input hashes also pass.

These identities validate implementation consistency, not demand identification or the assumption that observed prices are an equilibrium. The next step is the **manual damped pricing fixed point** from separate initial prices, followed by residual and unilateral-deviation checks.

Code: `1_static_demand_functions.R`, `2_recover_observed_market.R`, and `3_validate_observed_recovery.R` under `code/4_static_pricing_subgame/`. Outputs: incumbent recovery, fringe/market recovery, and interpretation. Solver-ready monthly inputs are saved as `observed_market_inputs.rds` in that output directory. Canonical panels and demand fits are unchanged.

4. Manual static pricing fixed point: observed-market replication

Completed October 28, 2026. Use the recovered October 27 costs and demand states from Section 3, with the full 540 incumbent rows in 132 monthly markets. Brands are vertically integrated single-product pricing firms. Incumbent quality, residuals, marginal costs, roster, and fringe total utility are fixed within each market; all incumbent and fringe shares are recomputed at each trial price vector.

Algorithm

Implement simultaneous damped markup substitution directly, without an equilibrium/root/optimization package:

$$T_i(p) = \widehat{mc}_i - \frac{1}{\widehat{\alpha} A_i(p)}, \qquad p_i^{(k+1)} = (1-\lambda)p_i^{(k)} + \lambda T_i(p^{(k)}),$$

$$A_i(p) = \frac{1}{1-\widehat{\sigma}} - \frac{\widehat{\sigma}}{1-\widehat{\sigma}} s_{i|g}(p) - s_i(p).$$

Four starting vectors: marginal cost + $0.10/mL, +$1/mL, +$5/mL, and heterogeneous positive cost increments. Cross these with damping values 0.25, 0.5 and 1: **12 attempts per month, 1,584 attempts total**. No observed price vector is used as a starting vector or convergence target. The reference solution is fixed in advance as cost+$1 with damping 0.5.

Require all three: maximum price step < 1e-10, maximum undamped map residual < 1e-9, and maximum scaled FOC residual < 1e-8. Cap at 2,000 iterations and report nonconvergence explicitly. The scaled FOC is `1 + (price - mc) * $\alpha$ * A`, avoiding dependence of the stopping rule on the volume units. Save every iteration's diagnostics. This is a markup fixed point, not an exact best-response iteration.

Results

All 1,584 attempts converged to the same observed-price solution within numerical tolerance. The largest observed-price discrepancy across all starts/damping values is below **3e-10 real dollars/mL**. The predeclared reference solutions differ from observed prices by less than 1e-10.

Damping	Attempts / converged	Median iterations	Maximum iterations
0.25	528 / 528	73	82

Damping	Attempts / converged	Median iterations	Maximum iterations
0.50	528 / 528	32	36
1.00 (undamped)	528 / 528	57	1,335

Damping 0.5 is the practical starting choice for these observed markets; it is not a general convergence guarantee for counterfactual markets.

Unilateral profit-deviation checks

For every one of the 540 reference incumbent solutions, hold rivals' prices fixed and find its best own price by **manually coded bracketing and bisection** on the own-profit FOC. This is separate from the multi-firm fixed-point solve.

For own price at or above marginal cost, `A_i > 0` and its own-price derivative is positive. Since $\alpha < 0$, `g_i = 1 + $\alpha$ *(p_i-mc_i)*A_i` is strictly decreasing: it starts at 1 at cost and becomes negative at sufficiently high price. Its unique zero maximizes own variable profit; below-cost profits are nonpositive and profit tends to zero at infinite price. Thus the manually bracketed root checks the global unilateral optimum under the maintained single-product demand model.

The largest best-response price gap is **9.81e-11 dollars/mL**. The maximum apparent profit gain is **4.48e-8 real dollars** (relative gain below **1.35e-14**), consistent with rounding error. Own-profit FOC slopes are negative for all 540 solutions. These checks support numerical Nash optimality for the maintained game; they do not establish uniqueness of the joint equilibrium.

Interpretation

This confirms that the manual algorithm can reproduce the observed-month equilibrium implied by the recovered demand states and marginal costs. Because costs were inverted from those observed prices, replication is an implementation check, not independent evidence that demand, integrated-channel conduct, instruments, or costs are empirically correct. No Markov process, fixed operating cost, entry/exit decision, or investment is required for this step. The replication exercise does not replace observed inputs or demand estimates; a separate numerical cost-perturbation check is described below.

Files and checks

Scripts 4–6 in `code/4_static_pricing_subgame/` implement and validate this step. Additional checks cover permutation invariance, the sigma-zero logit case, explicit failure under an insufficient iteration budget, probability mass balance, and unchanged recovery-input hashes; all pass.

Outputs: convergence summary, all solver attempts, monthly price solutions, unilateral deviation checks, and interpretation. Full traces and RDS results are in the same directory.

Visual audit and numerical cost perturbation

Created an audit plot booklet with observed/solved firm prices, separate replication-error time series, price/cost/markup series, convergence residuals for the first/middle/slowest months, iteration counts across all runs, and unilateral profit curves for every incumbent in those selected months. Each plot also has standalone PDF/PNG files and supporting curve CSVs in the same `audit/` folder. The profit maxima coincide with the solved prices. Script: `code/4_static_pricing_subgame/7_audit_plots.R`.

A separate numerical counterfactual raises VUSE's recovered marginal cost by 10% in December 2023 (the largest-volume firm in the latest observed month). Its cost increases by \$0.3344/mL and solved price by \$0.2155/mL, implying 64.4% pass-through in this scenario. JUUL's price rises by \$0.0191/mL and NJOY's by \$0.0146/mL; their costs are unchanged. VUSE's share falls while its rivals' shares increase. Three independent starting vectors converge to the same counterfactual solution, and manually bisected unilateral best responses agree. This checks behavior away

from the inverted observed-price vector; it is not estimated causal tax/cost pass-through or independent empirical validation. Original recovery and pricing results remain unchanged.

How to interpret the audits: what I have and have not shown

Discussion added November 2, 2026. The audits distinguish numerical solution, individual profit maximization, and empirical model validity.

- **Audit 1: Observed versus solved prices** — prices overlap, with an all-start maximum error below 3×10^{-10}/mL. The separate error plot exposes the tiny differences. This shows that the solver recovers the price vector implied by the inverted costs and demand states—an internal replication check.
- **Audit 2: Price, cost, and markup time series** — inspect brand-level levels, jumps, and price–cost gaps. All 540 recovered marginal costs are positive, but positivity alone does not validate the cost estimates.
- **Audit 3: Convergence from MC + \$1/mL** — map and scaled-FOC residuals approach zero from starting prices that do not use observed prices. Damping 0.5 is consistently faster in the difficult month. This supports numerical convergence to a stationary solution; iterations are computational updates, not firms learning over calendar time.

Pricing first-order condition (FOC) for Audit 3: each firm chooses its own price to maximize

$$\pi_i(p) = (p_i - mc_i)q_i(p) - FC_i,$$

holding rivals' prices, costs, and non-price demand states fixed. Its interior pricing condition is

$$\frac{\partial \pi_i}{\partial p_i} = q_i + (p_i - mc_i)\frac{\partial q_i}{\partial p_i} = 0.$$

Since $q_i = M_t s_i$, divide by market size to obtain

$$s_i + (p_i - mc_i)\frac{\partial s_i}{\partial p_i} = 0.$$

Our nested logit gives $\partial s_i/\partial p_i = \alpha s_i A_i(p)$, where

$$A_i(p) = \frac{1}{1-\sigma} - \frac{\sigma}{1-\sigma}s_{i|g}(p) - s_i(p).$$

Dividing by positive s_i gives the **scaled FOC used in the audit graphs**:

$$\boxed{1 + \alpha(p_i - mc_i)A_i(p) = 0.}$$

The plotted scaled-FOC residual is the maximum absolute value of this expression across firms at each iteration. Near zero means that the marginal benefit of a higher price balances the profit lost from lower sales. The separate unilateral-profit checks establish maximization under the maintained model; the FOC alone establishes stationarity.

- **Audit 4: Multiple starts and damping values** — all 1,584 attempts converge to the observed-price solution within tolerance. This establishes robustness to the tested starting points and numerical settings, not equilibrium uniqueness.
- **Audit 5: Unilateral profit curves and best responses** — holding rivals' prices fixed, solved prices coincide with own-profit maxima in the plotted months. Manually bisected checks cover all 540 firm-months and find no profitable deviation beyond rounding. This supports a numerical Nash equilibrium under the maintained single-product game, rather than merely zero first derivatives.

- **Audit 6: VUSE numerical cost perturbation** — a 10% cost increase in December 2023 raises its price by about \$0.216/mL and rivals' prices slightly. Independent starts and unilateral checks agree. This checks solver behavior away from observed prices under this scenario; it does not estimate causal real-world cost or tax pass-through.
- **Audit 7: Sequential markup updates (Gauss–Seidel)** — update one firm's price at a time using the newest price vector, then sweep through all firms again. Tested damping 0.5 and 1, four initial price vectors, and alphabetical/reverse brand orders for all 132 months. All 2,112 attempts converge to the simultaneous solution within numerical tolerance. Damping 0.5 requires at most 35 full sweeps; undamped updates at most 137.
- **Audit 8: Sequential best responses** — hold rivals' current prices fixed, find one firm's profit-maximizing price using manual bracketing/bisection, update it, then move to the next firm. All 1,056 attempts converge, across the same months, four starts and two orders, in at most 11 full sweeps (median 8). Each sweep includes inner bisection work; fewer sweeps do not necessarily mean less computational effort.

Across **all 3,168 sequential attempts**, the maximum price difference from the simultaneous solution is below **1.75e-10 real dollars/mL**. Final-vector map/FOC checks and unilateral best-response checks pass for every run; maximum relative profit gain is below **1.61e-14**, consistent with rounding. Residuals are checked for all firms after a complete sweep, since later updates can change earlier firms' optimal prices. Agreement supports numerical robustness to update method and the tested order/starts, not a proof of joint equilibrium uniqueness. These computational sequences still solve the same simultaneous-move economic game.

Sequential audit plots, results by method/order, and combined audit booklet. Code: `8_sequential_pricing_functions.R` and `9_audit_sequential_pricing.R` in `code/4_static_pricing_subgame/`. Original demand, recovery, and simultaneous-solve inputs are unchanged.

Why observed-price replication does not select the demand specification. Demand residuals were recovered to match observed shares, and costs were recovered by assuming observed prices satisfy the chosen model's Bertrand FOCs. Therefore observed prices are a solution by construction. Solving from other initial prices and checking profit maxima tests numerical implementation and compatibility with that maintained game. It does not independently identify the correct demand elasticity, nesting structure, or instrument set. A shared mistake in recovery and pricing could also cancel, which is why independently checked demand derivatives and unilateral best responses are useful.

A different logit/nested-logit specification could reproduce the **same observed prices and shares** and pass no-profitable-deviation checks by implying **different marginal costs**. It could nevertheless yield different markups, profits, and counterfactual responses. This is not guaranteed for every alternative: a specification could imply implausible costs or fail profit-maximization checks. Passing our checks does not establish that the chosen specification is preferable to all alternatives.

Maintained economic assumptions: simultaneous price competition; each brand acts as a vertically integrated single-product pricing firm; profit is $(p_i - mc_i)q_i - FC_i$ with constant marginal cost within a month; quantities respond according to the chosen nested logit; fringe utility is passive within the pricing game; and current prices do not affect future states or continuation values. The audits do not establish that actual firms behave this way or that recovered integrated-channel costs equal manufacturer production costs. They also do not resolve demand-instrument exclusion, the negative tax–price relationship, market-size validity, or uncertainty in the selected coefficients.

Summary

- I implemented the pricing fixed point manually. Across all observed months and multiple starting vectors, it reproduces observed prices to numerical precision and satisfies the pricing first-order conditions.
- Holding rivals' prices fixed, each solved price maximizes the specified firm's variable profit. The unilateral checks support a numerical Nash equilibrium under the maintained model, not just a stationary point.
- This is a computational consistency result, not validation of the selected nested-logit specification. Costs and demand residuals were recovered using the same observed prices and shares, so another demand specification could rationalize those observations with different costs and markups.
- I have established a working static pricing baseline. I have not established equilibrium uniqueness or empirical validity of the assumed demand, price competition, vertical integration, and fringe treatment.

Questions for Marc

[Add questions as the exercises develop.]

Meeting discussion and agreed next steps

[Populate after the meeting.]